

EXP 03 - Decision Tree

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, export_text
```

```
# Register Number
```

```
print("Register Number: 192321113\n")
```

```
# Loan Approval Dataset
```

```
data = {
```

```
    'Income': [
```

```
        'High','High','Medium','Low','Medium',
```

```
        'Low','High','Medium','Low','High',
```

```
        'Medium','High','Low','Medium','High',
```

```
        'Low','Medium','High','Low','Medium',
```

```
        'High','Low','Medium','High','Low',
```

```
        'Medium','High','Low','Medium','High'
```

```
    ],
```

```
    'CreditScore': [
```

```
        'Good','Average','Good','Poor','Average',
```

```
        'Average','Poor','Good','Good','Good',
```

```
        'Poor','Average','Poor','Good','Good',
```

```
        'Average','Poor','Good','Poor','Average',
```

```
        'Good','Poor','Average','Good','Poor',
```

```
        'Good','Average','Good','Poor','Good'
```

```
    ],
```

```
    'Employment': [
```

```
        'Yes','Yes','Yes','No','No',
```

```
        'Yes','Yes','No','No','No',
```

```
        'Yes','No','Yes','Yes','Yes',
```

```
        'No','No','Yes','Yes','No',
```

```
        'Yes','No','Yes','Yes','No',
```

```
        'Yes','Yes','No','Yes','No'
```

```
    ],
```

```
    'Property': [
```

```

        'Yes','Yes','No','No','Yes',
        'No','Yes','No','No','Yes',
        'Yes','No','Yes','Yes','No',
        'No','Yes','Yes','No','Yes',
        'Yes','No','Yes','No','Yes',
        'Yes','Yes','No','No','Yes'
    ],

    'Age': [
        'Young','Middle','Middle','Old','Young',
        'Old','Middle','Young','Old','Middle',
        'Young','Old','Middle','Young','Middle',
        'Old','Young','Middle','Old','Young',
        'Middle','Old','Young','Middle','Old',
        'Young','Middle','Old','Young','Middle'
    ],

    'Education': [
        'Graduate','Graduate','Graduate','UG','Graduate',
        'UG','Graduate','UG','UG','Graduate',
        'Graduate','UG','Graduate','Graduate','UG',
        'UG','Graduate','Graduate','UG','Graduate',
        'Graduate','UG','Graduate','Graduate','UG',
        'Graduate','Graduate','UG','Graduate','Graduate'
    ],

    'LoanApproved': [
        'Yes','Yes','Yes','No','No',
        'No','Yes','Yes','No','Yes',
        'Yes','No','No','Yes','Yes',
        'No','No','Yes','No','Yes',
        'Yes','No','Yes','Yes','No',
        'Yes','Yes','No','No','Yes'
    ]
}

# Create DataFrame
df = pd.DataFrame(data)

```

```

print("Loan Approval Dataset\n")
print(df)

# Separate input and output
X = df.iloc[:, :-1].copy()
y = df['LoanApproved']

# Encode categorical values
encoders = {}

for column in X.columns:
    le = LabelEncoder()
    X[column] = le.fit_transform(X[column])
    encoders[column] = le

target = LabelEncoder()
y = target.fit_transform(y)

# Build Decision Tree using ID3 (Entropy)
model = DecisionTreeClassifier(criterion='entropy', random_state=0)
model.fit(X, y)

# Display Decision Tree
print("\nDecision Tree\n")
print(export_text(model, feature_names=list(X.columns)))

# New Sample
new_sample = pd.DataFrame({
    'Income': ['Medium'],
    'CreditScore': ['Good'],
    'Employment': ['Yes'],
    'Property': ['Yes'],
    'Age': ['Young'],
    'Education': ['Graduate']
})

# Encode New Sample
for column in new_sample.columns:
    new_sample[column] = encoders[column].transform(new_sample[column])

```

```

# Prediction
prediction = model.predict(new_sample)

# Convert prediction back to Yes/No
result = target.inverse_transform(prediction)

print("\nNew Sample")
print("Income    = Medium")
print("CreditScore = Good")
print("Employment  = Yes")
print("Property    = Yes")
print("Age         = Young")
print("Education   = Graduate")

print("\nPrediction")
print("Loan Approved =", result[0])

```

Register Number: 192321113



*** Loan Approval Dataset

	Income	CreditScore	Employment	Property	Age	Education	LoanApproved
0	High	Good	Yes	Yes	Young	Graduate	Yes
1	High	Average	Yes	Yes	Middle	Graduate	Yes
2	Medium	Good	Yes	No	Middle	Graduate	Yes
3	Low	Poor	No	No	Old	UG	No
4	Medium	Average	No	Yes	Young	Graduate	No
5	Low	Average	Yes	No	Old	UG	No
6	High	Poor	Yes	Yes	Middle	Graduate	Yes
7	Medium	Good	No	No	Young	UG	Yes
8	Low	Good	No	No	Old	UG	No
9	High	Good	No	Yes	Middle	Graduate	Yes
10	Medium	Poor	Yes	Yes	Young	Graduate	Yes
11	High	Average	No	No	Old	UG	No
12	Low	Poor	Yes	Yes	Middle	Graduate	No
13	Medium	Good	Yes	Yes	Young	Graduate	Yes
14	High	Good	Yes	No	Middle	UG	Yes
15	Low	Average	No	No	Old	UG	No
16	Medium	Poor	No	Yes	Young	Graduate	No
17	High	Good	Yes	Yes	Middle	Graduate	Yes
18	Low	Poor	Yes	No	Old	UG	No
19	Medium	Average	No	Yes	Young	Graduate	Yes
20	High	Good	Yes	Yes	Middle	Graduate	Yes
21	Low	Poor	No	No	Old	UG	No

Activate Windows
Go to Settings to activate Windows.

21	Low	Poor	No	No	Old	UG	No
22	Medium	Average	Yes	Yes	Young	Graduate	Yes
23	High	Good	Yes	No	Middle	Graduate	Yes
24	Low	Poor	No	Yes	Old	UG	No
25	Medium	Good	Yes	Yes	Young	Graduate	Yes
26	High	Average	Yes	Yes	Middle	Graduate	Yes
27	Low	Good	No	No	Old	UG	No
28	Medium	Poor	Yes	No	Young	Graduate	No
29	High	Good	No	Yes	Middle	Graduate	Yes



Decision Tree

```

|--- Education <= 0.50
|   |--- Income <= 0.50
|   |   |--- class: 1
|   |   |--- Income > 0.50
|   |       |--- CreditScore <= 1.50
|   |       |   |--- Employment <= 0.50
|   |       |   |   |--- class: 0
|   |       |   |   |--- Employment > 0.50
|   |       |   |   |   |--- class: 1
|   |       |   |--- CreditScore > 1.50
|   |       |       |--- Age <= 1.00
|   |       |       |   |--- class: 0
|   |       |       |   |--- Age > 1.00
|   |       |       |       |--- Employment <= 0.50
|   |       |       |       |   |--- class: 0

```

Activate Windows
Go to Settings to activate Windows.

```

|   |   |   |   |   |--- Property <= 0.50
|   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |--- Property > 0.50
|   |   |   |   |   |   |--- class: 1
|--- Education > 0.50
|   |--- Age <= 0.50
|   |   |--- class: 1
|   |--- Age > 0.50
|   |   |--- Income <= 1.50
|   |   |   |--- class: 0
|   |   |--- Income > 1.50
|   |       |--- class: 1

```



New Sample

Income = Medium
CreditScore = Good
Employment = Yes
Property = Yes
Age = Young
Education = Graduate

Prediction

Loan Approved = Yes